EXERCISE SHEET 5

GROUP AND RING THEORY

Main questions - to be done before the seminar

5.1. (*Modular law*) Let A, B and C be R-submodules of an R-module M such that $A \supseteq B$. Show that

$$A \cap (B + C) = B + (A \cap C).$$

Give an example of an R-module M and three R-submodules A, B and C of M such that

$$A \cap (B + C) \neq (A \cap B) + (A \cap C).$$

In other words the set of all R-submodules of an R-module M is a module lattice which is not necessarily distributive.

5.2. Let R be a ring. Recall that an *idempotent* is an element $a \in R$ such that $a^2 = a$. For any two idempotents e, f in R, prove that

$$Re + Rf = Re \oplus R(f - fe).$$

5.3. (*Second Isomorphism Theorem*) If A and B are R-submodules of an R-module M, show that $(A + B)/A \cong B/(A \cap B)$ as R-modules.

Extra questions

5.4. If $f : M \to N$ is an R-homomorphism of an R-module M to an R-module N, show that

(a) $\ker f$ is an R-submodule of M;
(b) $\operatorname{im} f$ is an R-submodule of N.

5.5. Let R be a ring. Show that the R-isomorphism is an equivalence relation in the set of R-modules.